

Paragon: Open Source in Crisis Management

Lessons Learned on Using Open Source and Open Data in Developing
Crisis Management Tools

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Qui suis-je ?



1

Background

Gael Kruwialis, geographer,
convinced by administration at
the service of the public ...

2

Experience

Worked with open source for
Brussels region, IGN, and
NCCN

3

Focus

Implementing user services
and organize data for crisis
management solutions

Contexte :

Le Centre de crise National

Anticiper pour mieux gérer



LES ATTENTATS

11/09/2001



FUKUSHIMA

11/03/2011



PUKKELPOP

18/08/2011



CATASTROPHE FERROVIAIRE

04/05/13

ANTICIPER POUR MIEUX GÉRER

Cycle du Risque National



MISSIONS

- 1.Strengthen the resilience of our society
- 2.Organize emergency planning and crisis management at the national level
- 3.Ensure active vigilance
- 4.Processing of passenger data

Why open source in crisis management



- Belgian and European context
- Importance of interoperability and technological independence
- Significant cost of proprietary licenses and independence
- Transparency: auditable code = trust
- Collaboration: sharing with other regions

PARAGON: An Open Source Platform for Crisis Management

- **Data:** Aggregation of open sources (OSM) and official sources (IGN).
- **Technology:** Open source stack (PostGIS, Apache Airflow, etc.).
- **Community:** Collaboration with OSGeo.be, public and private stakeholders.



Open source
based



Webbrowser



Multilingual



Responsive design



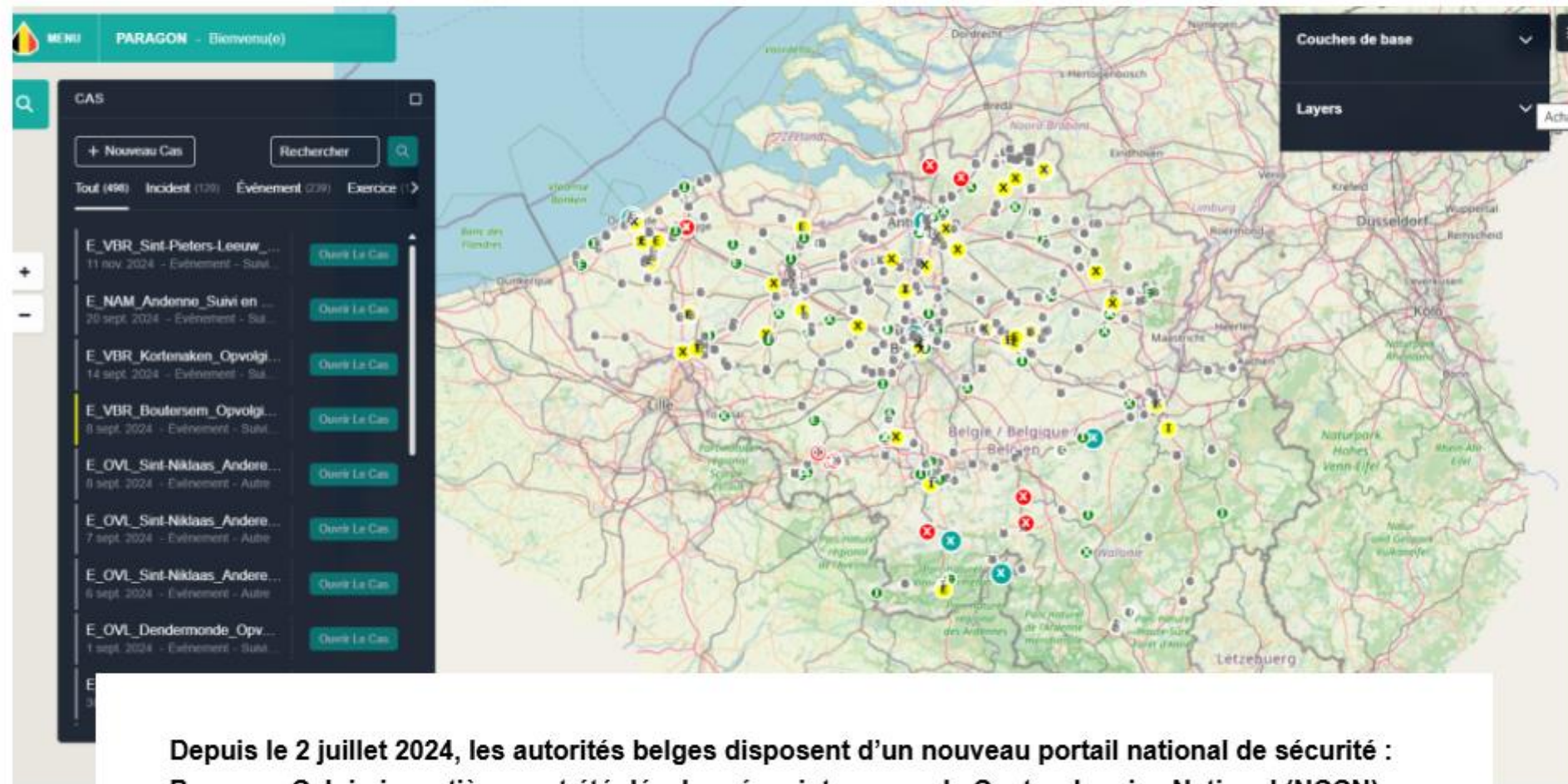
Scalable



Datacenter

Le Centre de crise National lance un outil innovant de planification d'urgence et de gestion de crise

02 août 2024



What is Paragon?

The screenshot displays the Paragon application interface, which is a map-based emergency response tool. The central map shows the Oud-Heverlee area, with various locations marked by colored icons (blue, orange, red). Two specific locations are highlighted with callout boxes: "Ferm Kinderopvang - Gezinsopvang Leys Rita nursery" and "Wijk Oud-Heverlee Commissariat de police".

The interface includes several panels and controls:

- Top Bar:** Contains a "DEV" button, a "MENU" button, a title bar "I VBR OUD-HEVERLEE PANIEK IN EEN MENIGTE SD", and a "General theme" dropdown.
- Left Panel:** Includes a vertical toolbar with icons for information, search, map, and other functions. Below the toolbar is a "FEED" section with a "MULTI" filter, a search bar, and a list of messages. The first message is from "Expert / Sven Duytschaever" dated "27 May 2025 10:37" with the text "hallo". Below the feed is an "Add post" section with radio buttons for "Neutral", "Important", and "Question", and a text input field "Type a new message". At the bottom of the left panel is an "EMERGENCY PLANS" section showing "0 Emergency Plan" and a search bar.
- Right Panel:** Contains three sections: "WEATHER Oud-Heverlee" showing "15°C", "0%", and "15 km/h"; "ANCHORS" showing "2 Anchors" and a search bar; and "CONTACTS" showing "Case Participants (25)", "Persons (0)", and "Organizations (0)".

The map itself shows a street grid with labels for "Oud-Heverlee", "Korbe Dijle", "Neerijse", "Doode Bemde", "Kouterbos", "Vaalseek", "Bl. len", "Tomberg 103 m", "Nationale Park Brabantse Wouden", and "Margijsbos". A scale bar indicates "1000 m".

PARAGON

Paragon key objects



Anchors



Contacts



Maps



Data



Plans



Feed



Case



Logistics



IOT/T&T/AI



Training



Flow



Analysis

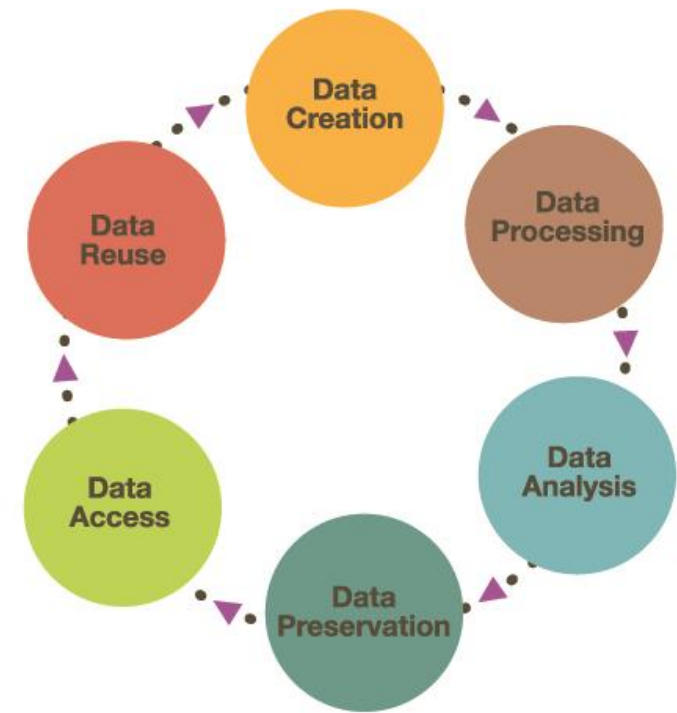
The screenshot displays the Paragon application interface. On the left, a sidebar menu contains two main sections: 'Achtergrondlagen' (Background layers) and 'Thematische kaartlagen' (Thematic map layers). The 'Achtergrondlagen' section includes options like 'Basiskaarten van de regio's', 'Luchtfoto's' (selected), 'NGI CartoWeb', 'NGI CartoWeb - greyscale', 'OpenFreeMap greyscale', 'OpenFreeMap positron', 'OpenStreetMap', 'OpenStreetMap Greyscale', and 'OpenTopoMap (Hoogte)'. The 'Thematische kaartlagen' section includes 'Afschakelplan', 'Bevolking', 'Case users', 'Cases' (checked), 'Lage- en middendruk gasleidingen (Vlaanderen)', 'Kust' (15), 'Oriëntatiehulp' (4), and 'Anker' (20). Below these, a 'Selecteer alles' link is present, followed by a list of thematic layers: 'Brandweerkazernes' (checked), 'Centra voor asielzoekers', 'Civiele bescherming', 'Cultureel Centrum', 'Fetrapi pijpleidingen', 'Gebedshuizen', 'Gevangenis', 'Internationale & militaire luchthavens' (checked), 'Internationale instellingen', 'Kinderopvang', 'Museum', and 'Politiebureaus' (checked). At the bottom of the sidebar, the logos for 'Nationaal Crisiscentrum' and 'Paragon' are visible. The main map area shows an aerial view of Brussels, with an orange outline highlighting a specific region. A tooltip for 'Brussels Military Airport' is visible, indicating it is a 'military airbase' at 'Level 2'. Other labels on the map include 'Melsbroek', 'Luchthavenbrandweer Post Noord', 'Brussels Airport', 'Luchthavenbrandweer Post West', 'Steenokkerzeel', 'HUMELGEM', 'Kortenberg', 'Leuven', 'Nossegem', 'Zaventem', and 'Politie Zaventem'. A scale bar at the bottom indicates 500 meters.

Everything that has a location and a name can be turned into an Anchor

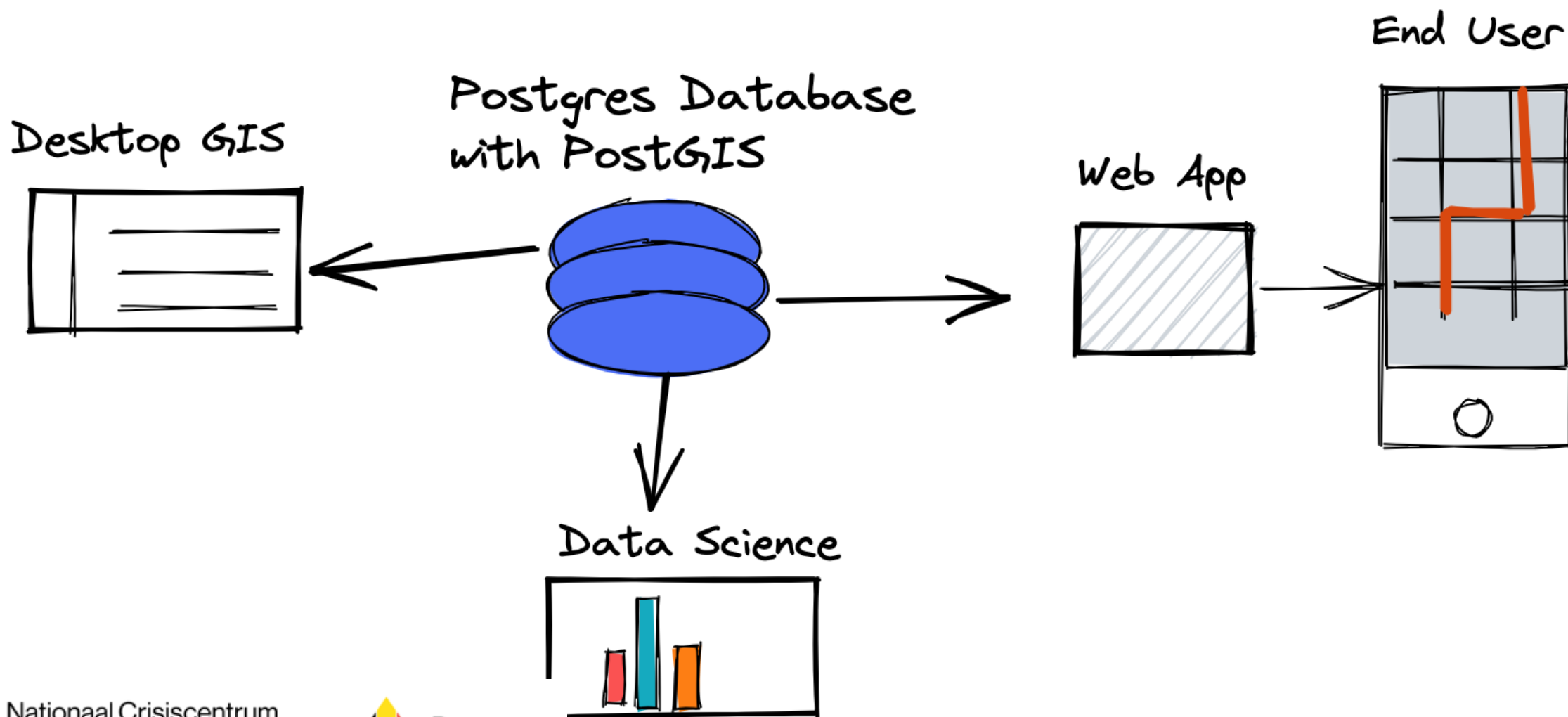
Strategy Geospatial Lakehouse

Too many layers & too many sources

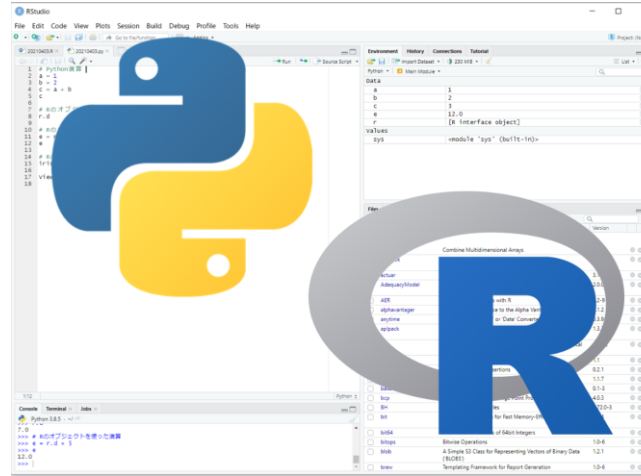
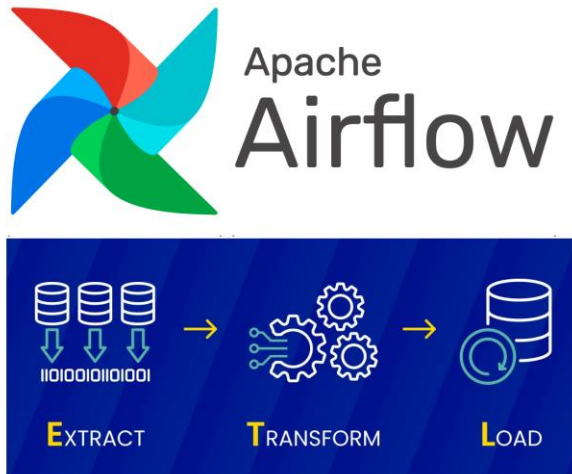
1. Focus on source stable and official
2. Categorization of layers
3. Optimisation of the visualisation
(Graphic semiology)
4. Consolidation (multiple source)
5. Data enhancement
6. Metadata monitoring and linked data



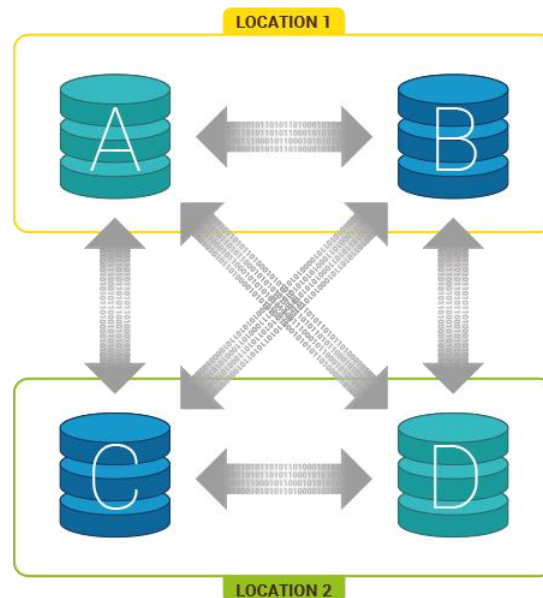
Acteurs



Workflow



Azure Database for PostgreSQL...



Proto-anchor planning & overview

Layers in production

existing layer	status/timing	in anchor search	in layer widget	update cycle	used sources
Addresses / Adressen / Adresses	ready	yes	no	continuously	
Administrative boundaries / Administratieve grenzen / Limites administratives	ready	yes	no	as needed	
Hospitals / Ziekenhuizen / Hôpitaux	ready	yes	yes	(monthly)	OSM (planned to be extended with CoBRHA)
Schools / Scholen / Écoles	ready	yes	yes	(monthly)	OSM + community datasets
Seveso companies / Seveso-bedrijven / Entreprises Seveso	ready	yes	yes	monthly	
Nuclear installations / Nucleaire installaties / Installations nucléaires	ready	yes	yes	on demand	
Pipelines / Pijpleidingen / Pipelines	ready	yes	yes	+/- monthly	
Population figures for estimating the number of people in a perimeter / Inwonersaantallen zodat er een inschatting kan komen van het aantal mensen in een perimeter / Nombre d'habitants pour estimer le nombre de personnes dans une zone donnée	ready	yes	yes	yearly	statbel
Police stations / Politieposten / Postes de police	ready	yes	yes	(monthly)/yearly	OSM+fedpol
Fire stations / Brandweerposten / Casernes de pompiers	ready	yes	yes	(monthly)	OSM (fixed with federal list)
Airports / Luchthavens / Aéroports	ready	yes	yes	on request	OSM
Nursing homes / Rusthuizen / Maisons de retraite	ready	yes	yes	can be continuously	CoBRHA + OSM + Sociale Kaart
Prisons / Gevangenen / Prisons	ready	yes	yes	can be continuously	OSM
Fuel storage and other industries / Brandstofopslag en andere industrie / Stockage de carburant et autres industries	NA	NA	NA		
Seaports / Zeehavens / Ports maritimes	ready	yes	no	on request	Vlaio
Cultural centers, cinemas, theaters, concert halls / Culturele centra, bioscopen, theaters, concertzalen / Centres culturels, cinémas, théâtres, salles de concert	ready	yes	yes	(monthly)	OSM (maybe + regional datasets)
Sports infrastructure / Sportinfrastructuur / Infrastructures sportives	ready	yes	yes	(monthly)	OSM
Religious buildings / Religieuze gebouwen / Bâtiments religieux	ready	!	!	!	!
Train stations / Treinstations / Gares ferroviaires	ready	!	!	!	!

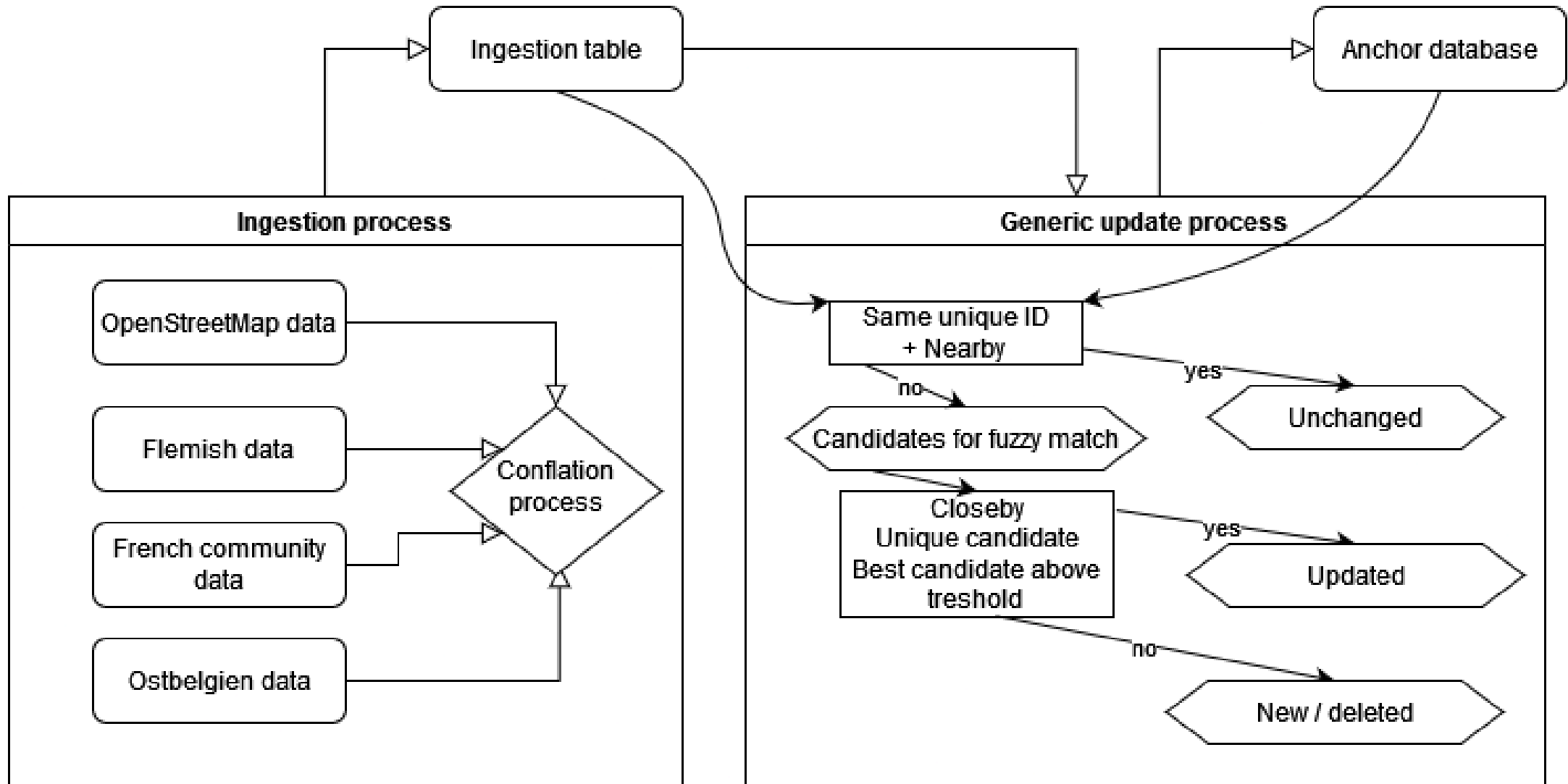


Nationaal Crisiscentrum
Centre de crise National



Paragon

UPDATE PROCESS



Gouvernance & transfert de compétences

- **Challenges: Resistance to change** => **Solution:** Training and documentation
- **Challenges: Integration of sensitive data** => **Solution:** Involvement of public stakeholders => **Solution:** Governance sharing with community and private sector
- **Challenges: Updates** => **Solution:** Ensure sustainable autonomy
=> **Solution:** Automation



Lessons Learned

- **Strengths:**
 - Rapid implementation
 - Richness of the open ecosystem
 - Cooperation among stakeholders
- **Challenges:**
 - Skills management
 - Sustainability
 - Data harmonization

Conclusion

- **Open source as a catalyst for cooperation between institutions and communities.**
- **A lever for a sovereign digital ecosystem.**
- **Open source strengthens resilience.**

Resilience is built together—**open source can save lives.**

